

DSA STUDY BLUEPRINT SERIES

115. Flood Fill Algorithm Graph Problem

Recoloring Connected Pixels in Grid (Flood Fill)

Section 10 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Problem:** Recolour connected pixels matching an initial colour.
- Run DFS/BFS starting at target pixel, updating matching neighbors.

Memory & Execution Blueprint



C++ Implementation Reference

Standard code layout and syntax

```
void dfsFill(int r, int c, vector>& img, int oldColor, int newColor) {  
    if (r<0||c<0||r>=img.size()||c>=img[0].size()||img[r][c]!=oldColor) return;  
    img[r][c] = newColor;  
    dfsFill(r+1, c, img, oldColor, newColor); dfsFill(r-1, c, img, oldColor, newColor);  
    dfsFill(r, c+1, img, oldColor, newColor); dfsFill(r, c-1, img, oldColor, newColor);  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(R * C)$
Space Complexity	$O(R * C)$ stack

Key Practices & Gotchas

- **Important:** Avoid infinite loops by checking that the new color differs from the old color.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace Dijkstra distance relaxation updates on active vertices:

Vertex popped	Adjacent edge checked	Current distance value	New path calculation	Relaxation action
U (Dist = 2)	U → V (Weight = 3)	V (Dist = 7)	New Dist = 2 + 3 = 5	Relax: Update V (Dist = 5)

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Shortest Path choices:** BFS finds unweighted short paths; Dijkstra handles positive weights; Bellman-Ford resolves negative edges.
- **Spanning Trees:** Prim's builds MST from a start node; Kruskal's sorts all edges using disjoint sets.
- **Related LeetCode Problems:** #200 Number of Islands, #743 Network Delay Time, #787 Cheapest Flights Within K Stops.